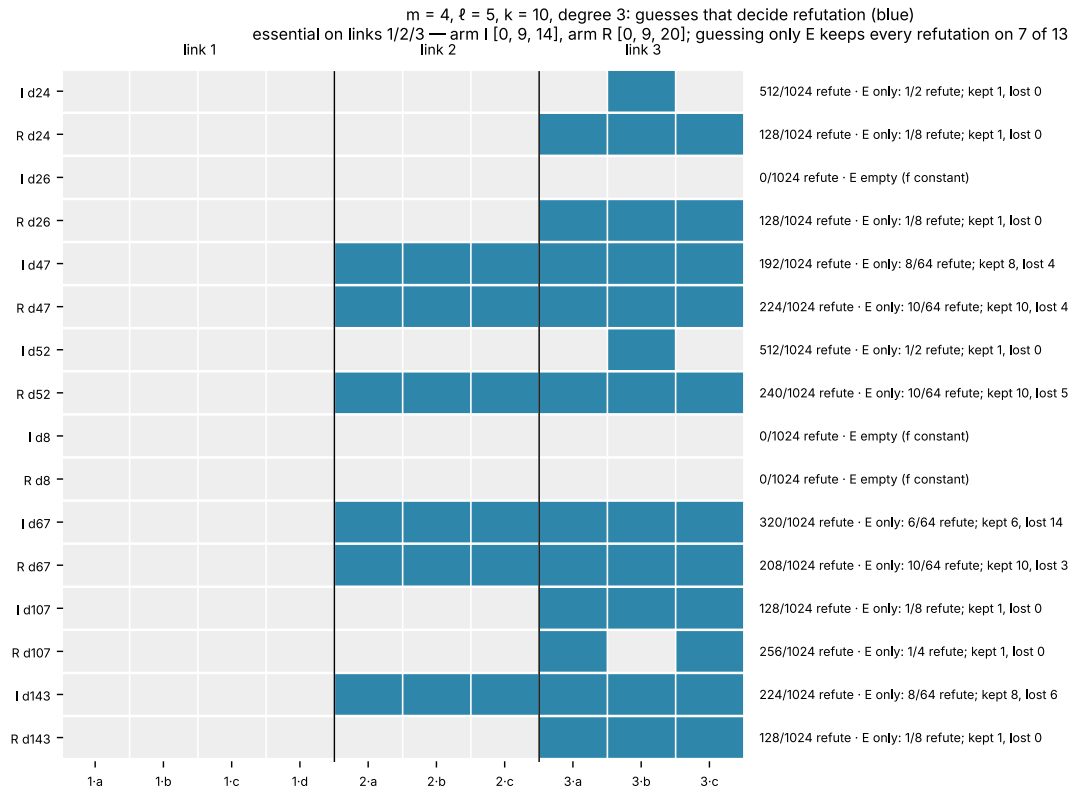
At m = 4 the link-1 guesses never decide degree-3 refutation

Study `research/dreg_m4_guess_bits_20261106/`, pre-registered. **Class:** stage diagnostic. **Standing:** unverified assertion until a pcc verdict exists.

Question

In 3571 every per-draw refuting count at m = 4 (ten guesses, degree 3) was a multiple of 16. Which of the ten guesses decide refutation, and does guessing only those keep the refutations?

Result



- No link-1 guess (pair sums in arm I, raw x_2 in arm R) is essential on any of the 16 arm-draws. Essential guesses sit on link 3 (34) and link 2 (18).
- Guessing only the essential set keeps every refutation on 7 of 13 tested arm-draws, all with one to three guesses; on arm I draws 24 and 52, one $e_2 + x_4$ guess (2 branches) keeps the refuting assignment that ten guesses (1024 branches) found.
- Controls valid, and the per-draw counts reproduce 3571

Reading

- At m = 4, degree 3, the work is done by guesses on the last links, the opposite of m = 3 at degree 4, where the pair invariant was the useful guess.
- A guess budget at m = 4 should go to links 2 and 3 first.
- No row moves, and rho is not reached.